

A Half-Parameter Partial Answer to the Fixed-Theta Tsirelson Depth Problem

Automatic math-discovery packet

June 29, 2026

Source Question

The source paper is Jakub Hodor, *The depth of Tsirelson's norm*, arXiv:2306.10344. It proves that the classical case $T[\frac{1}{2}, \mathcal{S}_1]$ has depth function $j(n) = \Theta(\sqrt{n})$, and then asks for the corresponding order when the parameter θ varies.

By Theorem 1, we know that if the constant C exists, then $C \in [\sqrt{2}, 2]$.

Recall that a generalized Tsirelson's norm $T[\theta, \mathcal{F}]$ depends on a real number $0 < \theta < 1$ and a regular family $\mathcal{F}$. In this paper, we studied the function $j_{T[\frac{1}{2}, \mathcal{F}]}$ for some regular families $\mathcal{F}$. Another approach is to determine what is the order of magnitude of the function $j_{T[\theta, \mathcal{F}]}$ when we fix $\mathcal{F} = \mathcal{S}_1$ and change θ . There are two versions of this problem.

Problem 2. For a fixed real number θ with $0 < \theta < 1$, compute the order of magnitude of the function $j_{T[\theta, \mathcal{S}_1]}(n)$.

Problem 3. Compute the order of magnitude of a two-variable function $j_{T[\theta, \mathcal{S}_1]}(n)$.

Figure 1: The fixed- θ and two-variable open problems in Hodor's paper, page 24.

Definitions

Let $0 < \theta < 1$ and let $\mathcal{F}$ be a regular family of finite subsets of $\mathbb{N}$. Put

$$W_0^\theta(\mathcal{F}) = \{\pm e_i^* : i \in \mathbb{N}\}.$$

Given $W_m^\theta(\mathcal{F})$, define $W_{m+1}^\theta(\mathcal{F})$ to be

$$W_m^\theta(\mathcal{F}) \cup \left\{ \theta \sum_{i=1}^r f_i : f_i \in W_m^\theta(\mathcal{F}), \text{supp } f_1 < \dots < \text{supp } f_r, \{\min \text{supp } f_i : 1 \leq i \leq r\} \in \mathcal{F} \right\}.$$

Let $W^\theta(\mathcal{F}) = \bigcup_m W_m^\theta(\mathcal{F})$, define

$$\|x\|_{\theta, \mathcal{F}} = \sup\{f(x) : f \in W^\theta(\mathcal{F})\},$$

and let

$$j_{\theta, \mathcal{F}}(x) = \min\{\text{depth}_\theta(f) : f \in W^\theta(\mathcal{F}), f(x) = \|x\|_{\theta, \mathcal{F}}\}.$$

Finally,

$$j_{\theta, \mathcal{F}}(a, b) = \max\{j_{\theta, \mathcal{F}}(x) : \text{supp } x \subset [a, b]\}, \quad j_{\theta, \mathcal{F}}(n) = j_{\theta, \mathcal{F}}(1, n).$$

For $\mathcal{F} = \mathcal{S}_1 = \{E \subset \mathbb{N} : |E| \leq \min E\}$, this is the function denoted $j_{T[\theta, \mathcal{S}_1]}(n)$ in the source problem.

Claim

Theorem 1. *For every fixed $0 < \theta \leq \frac{1}{2}$,*

$$j_{T[\theta, \mathcal{S}_1]}(n) = \Theta_\theta(\sqrt{n}).$$

More explicitly, the upper bound is $j_{T[\theta, \mathcal{S}_1]}(n) = O(\sqrt{n})$ with the same asymptotic constant supplied by Hodor's recursive estimate, and the lower bound is $j_{T[\theta, \mathcal{S}_1]}(n) = \Omega_\theta(\sqrt{n})$.

Proof Intuition

Hodor's lower bound for $\theta = \frac{1}{2}$ works by placing very large coordinates at the outside of a nested chain of full Schreier sets. The same forcing works for general fixed θ , provided the outer full Schreier sets have size $> 1/\theta + 1$ and the large coordinates are separated by a factor depending on θ .

The upper bound is more delicate. Most of Hodor's upper-bound proof is combinatorial, but one insertion lemma uses the identity that, at $\theta = \frac{1}{2}$, a certain norming functional is the average of two shallower competitors. For $0 < \theta \leq \frac{1}{2}$, the same functional is a convex combination of three competitors with weights $(1 - 2\theta), \theta, \theta$. This is exactly where the restriction $\theta \leq \frac{1}{2}$ enters.

Lower Bound

We use the following theta-version of the source paper's lower-bound forcing lemma.

Lemma 1 (Theta-forcing lower lemma). *Let $0 < \theta < 1$, let $\mathcal{F}$ be a regular family, and let $d \in \mathbb{N}$. Assume that there are $F_1, \dots, F_d \in \text{full}(\mathcal{F})$ and intervals $[a_i, b_i - 1]$, $1 \leq i < d$, satisfying Hodor's nesting conditions:*

1. a_i, b_i are consecutive in F_i ;

2. $a_i \in F_{i+1}$ and $F_{i+1} \subset [a_i, b_i - 1]$;
3. for distinct $u, v \in [a_i, b_i - 1]$,

$$(F_i \setminus \{a_i\}) \cup \{u, v\} \notin \mathcal{F}.$$

Assume also that

$$|F_i| - 1 > \frac{1}{\theta} \quad (1 \leq i \leq d).$$

Then there is $x \in c_{00}^+$, supported by $\bigcup_i F_i$, such that

$$j_{\theta, \mathcal{F}}(x) \geq d.$$

Proof. Choose once and for all $L > 1/(1 - \theta)$. The proof is by induction on d . For $d = 1$, put $x_i = 1$ on F_1 and $x_i = 0$ otherwise. Since $\theta|F_1| > 1$, the functional $\theta \sum_{i \in F_1} e_i^*$ beats every coordinate functional, so no depth-zero functional norms x . Hence $j_{\theta, \mathcal{F}}(x) \geq 1$.

Assume the result for $d - 1$, applied to $F_2, \dots, F_d$. Obtain $x' \geq 0$, supported by $F' = \bigcup_{i=2}^d F_i$, such that $j_{\theta, \mathcal{F}}(x') \geq d - 1$. Let $s = \sum_i x'_i$. Define

$$x_i = \begin{cases} x'_i, & i \in F', \\ Ls, & i \in F_1 \setminus F', \\ 0, & \text{otherwise.} \end{cases}$$

Let f' norm x' , and set

$$g = \theta \left(f' + \sum_{i \in F_1 \setminus F'} e_i^* \right).$$

The nesting assumptions make $g \in W^\theta(\mathcal{F})$. Since $|F_1 \setminus F'| = |F_1| - 1$,

$$g(x) = \theta \|x'\|_{\theta, \mathcal{F}} + \theta(|F_1| - 1)Ls > Ls.$$

Thus a norming functional f for x cannot have depth zero.

Every positive-depth functional has point coefficients at most θ . If for some $i_0 \in F_1 \setminus F'$ the coefficient $f(e_{i_0})$ were strictly less than θ , then, because coefficients in a norming tree are powers of θ , we would have $f(e_{i_0}) \leq \theta^2$. Therefore

$$\begin{aligned} f(x) &\leq \theta \sum_{i \in (F_1 \cup F') \setminus \{i_0\}} x_i + \theta^2 x_{i_0} \\ &\leq \theta(|F_1| - 2)Ls + \theta s + \theta^2 Ls. \end{aligned}$$

Because $L > 1/(1 - \theta)$, the right side is strictly smaller than $\theta(|F_1| - 1)Ls \leq g(x)$, contradicting $f(x) = \|x\| \geq g(x)$. Hence $f(e_i) = \theta$ for every $i \in F_1 \setminus F'$.

It follows that

$$f = \theta \left(\sum_{\ell=1}^m h_\ell + \sum_{i \in F_1 \setminus F'} e_i^* \right)$$

for some block sequence $h_1, \dots, h_m$. The third nesting condition implies that at most one of the h_ℓ 's can start inside $[a_1, b_1 - 1]$. Comparing $f(x)$ with $g(x)$ gives

$$\sum_{\ell=1}^m h_\ell(x') \geq \|x'\|_{\theta, \mathcal{F}}.$$

All summands not starting in $[a_1, b_1 - 1]$ vanish on x' , and every summand is bounded above by $\|x'\|_{\theta, \mathcal{F}}$. Hence one summand h_ℓ norms x' . Since $j_{\theta, \mathcal{F}}(x') \geq d - 1$, we get $\text{depth}_\theta(h_\ell) \geq d - 1$, and so $\text{depth}_\theta(f) \geq d$. $\square$

Now apply the lemma to $\mathcal{F} = \mathcal{S}_1$. Let

$$q_\theta = \left\lfloor \frac{1}{\theta} \right\rfloor + 2.$$

For a given d , define integers $B_{d+1} = q_\theta + d + 1$ and

$$B_i = B_{i+1} + q_\theta + i - 3 \quad (1 \leq i \leq d).$$

Set

$$F_i = \{q_\theta + i - 1, q_\theta + i\} \cup [B_{i+1}, B_i - 1].$$

Then F_i is a full Schreier set: its cardinality is $q_\theta + i - 1 = \min F_i$. The sets satisfy the nesting hypotheses above, and $|F_i| - 1 > 1/\theta$. Therefore

$$j_{\theta, \mathcal{S}_1}(N_d) \geq d,$$

where

$$N_d = \max F_1 = q_\theta + d(q_\theta - 2) + \frac{d(d+1)}{2}.$$

This proves $j_{\theta, \mathcal{S}_1}(n) = \Omega_\theta(\sqrt{n})$.

Upper Bound for $0 < \theta \leq 1/2$

We recall only the coefficient-sensitive part of Hodor's upper-bound proof and explain its theta-version. The remaining steps use the same full-set and Schreier-family combinatorics as in the source paper.

The easy replacement lemmas remain valid with $\frac{1}{2}$ replaced by θ : if a positive-depth functional f has support in $\mathcal{F}$, then $\theta \sum_{i \in \text{supp } f} e_i^*$ is not worse than f on positive vectors; and a norming functional can be replaced by one that is either shallow or $\mathcal{F}$ -full.

The insertion lemma is the only place where the value $\frac{1}{2}$ is special. Suppose $\mathcal{F}$ has the insertion property, g' is a norming full functional chosen with maximal possible minimum support, and

$$g' = \theta(g_1 + \cdots + g_r), \quad g_1 = \theta(t_1 + \cdots + t_s),$$

with t_1 still too deep. Hodor forms two inserted competitors and uses an average identity. For $0 < \theta \leq \frac{1}{2}$, define

$$\begin{aligned} u_0 &= \theta(g_2 + \cdots + g_r), \\ u_1 &= \theta(t_1 + g_2 + \cdots + g_r), \\ u_2 &= \theta(t_2 + \cdots + t_s + g_2 + \cdots + g_r). \end{aligned}$$

Empty sums are simply omitted; the degenerate one-branch cases are easier, because a useless scalar branch can be replaced by its child. The insertion property and heredity of $\mathcal{F}$ make these functionals admissible. A direct coefficient check gives

$$g' = (1 - 2\theta)u_0 + \theta u_1 + \theta u_2.$$

Since g' norms x , and each $u_i(x) \leq \|x\|_{\theta, \mathcal{F}}$, every competitor with positive coefficient in this convex combination also norms x . In particular, u_2 norms x , has no larger depth than g' , and has strictly larger minimum support. Applying the shallow-or-full replacement lemma to u_2 contradicts the choice of g' . Thus Hodor's strengthened insertion lemma remains true for $0 < \theta \leq \frac{1}{2}$.

Consequently the realizing-sequence lemma and the recursive estimate for $\mathcal{S}_\varphi$ hold verbatim in this parameter range. For $\varphi(a) = a$, i.e. $\mathcal{S}_\varphi = \mathcal{S}_1$, one obtains: if $a \leq b$ and $\|[a, b]\| \leq a$, then

$$j_{\theta, \mathcal{S}_1}(a, b) \leq 1,$$

and otherwise

$$j_{\theta, S_1}(a, b) \leq 2 + \max\{j_{\theta, S_1}(a + i, b - a + i + 1) : 1 \leq i < a\}.$$

The same induction as in Hodor’s Section 7.1 yields

$$n \leq \sum_{i=1}^{c+1} i - c \implies j_{\theta, S_1}(n) \leq 2c + 1.$$

Choosing $c = O(\sqrt{n})$ proves

$$j_{\theta, S_1}(n) = O(\sqrt{n})$$

for every $0 < \theta \leq \frac{1}{2}$.

Combining this with the lower bound completes the proof of the theorem.

Why This Is Only Partial

The modified insertion step uses a convex combination with coefficient $1 - 2\theta$. When $\theta > \frac{1}{2}$, this coefficient is negative, and the argument no longer proves that a later-support competitor also norms the vector. A tempting alternative is to use linear-depth chain functionals on supports of size $O(d)$, but finite-dimensional LP checks for small supports show that these chains do not uniformly beat all shallower functionals except in very small or very large- θ toy cases. No proof-grade replacement for the insertion lemma in the range $\theta > \frac{1}{2}$ is included here.

Literature and Novelty Check

Before promotion, the cheap run indexes were searched for arXiv 2306.10344, for **depth of Tsirelson**, and for the exact theta-Schreier keywords; no prior run packet for this target was found. A bounded web search on June 29, 2026 used the phrases:

- `j_{T[theta, S_1]}`,
- `T[theta, S_1] j Tsirelson depth`,
- `The depth of Tsirelson’s norm theta Schreier`.

The search found the source paper and no later exact answer to the $\theta \leq \frac{1}{2}$ subcase. Novelty confidence is therefore bounded rather than absolute.

Verification Notes

The proof is not computational. The included script `code/theta_chain_lp_probe.py` records only a finite LP diagnostic for the unpromoted $\theta > \frac{1}{2}$ chain idea. The main human-review focus should be the two analytic modifications: Lemma 1 and the convex-combination replacement in the insertion lemma.

References

- [1] J. Hodor, *The depth of Tsirelson’s norm*, arXiv:2306.10344, 2023.